Let's learn.

Reading of Pie diagram

The following example illustrates how a pie chart gives information at a glance. 120 students of standard 10 were asked which game they like. The information obtained is shown in the adjacent pie diagram. Answers to the question as-

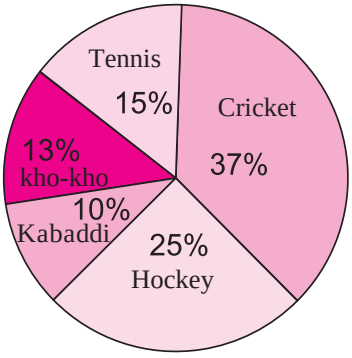

fig 6.9

'Which game is liked the most'

'What percentage of students like kho-kho?'

'What percentage of students like kabaddi?'

can be obtained from the pie diagram at a glance.

Observe one more pie diagram.

Figure 6.10 shows the annual financial planning of a school. From the pie diagram we see that

- 45% of the amount is reserved for educational equipment.
- 35% of the amount is shown for games.
- 10% of the amount is kept for sanitation.
- 10% of the amount is reserved for environment.

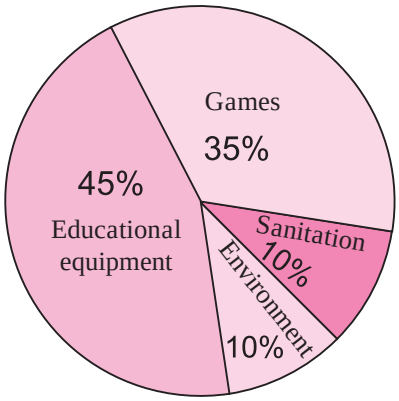

fig 6.10

In this way, we get information at a glance from a pie diagram.

Let us have more information about a pie diagram.

Many times we find information of different types in newspapers given in the form of pie diagrams. For example, the annual budget, performance of different nations in olympic games, etc.

Now we shall see, by examples, how to interpret the information from a pie diagram.

Example :

As deduced from a survey, the classification of skilled workers is shown in the pie diagram (fig 6.11). If the number of workers in the production sector is 4500, answer the following questions.

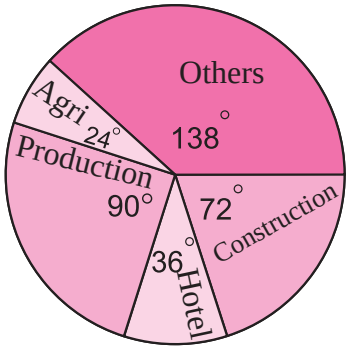

fig 6.11

(i) What is the total number of skilled workers in all fields ?

(ii) What is the number of skilled workers in the field of constructions ?

(iii) How many skilled workers are in agriculture ?

(iv) Find the difference between the numbers of workers in the field of production and construction.

Solution : (i) Suppose, the total number of skilled workers in all fields is x.

$\therefore$ the central angle for x persons is = 360°

Central angle for number of persons in production field

$$= \frac{\text{Number of persons in production field}}{x} \times 360$$

$$\therefore 90 = \frac{4500}{x} \times 360$$

$$\therefore x = 18000$$

$\therefore$ total number of skilled workers in all the fields together = 18000.

(ii) The angle shown for construction sector = 72°.

$$\therefore 72 = \frac{\text{Number of persons in construction}}{18000} \times 360$$

$$\therefore \text{number of persons in construction field} = \frac{72 \times 18000}{360}$$

$$= 3600$$

(iii) The central angle for agriculture field is 24°.

$$\therefore 24 = \frac{\text{Number of workers in agriculture}}{\text{total skilled workers}} \times 360$$

$$24 = \frac{\text{Number of workers in agriculculture}}{18000} \times 360$$

$$\therefore \text{number of workers in agriculture} = \frac{24 \times 18000}{360}$$

$$= 1200$$

(iv) The difference between angles relating fields of production and construction

$$= 90^\circ - 72^\circ = 18^\circ.$$

$\therefore$ The difference between the central angles =

$$\frac{\text{Difference between numbres of workers in the fields}}{\text{Total number of skilled workers}} \times 360$$

$$18 = \frac{\text{Difference between the numbers of workers in the fields}}{18000} \times 360$$

$$\text{Difference between the numbers of workers in the two fields} = \frac{18 \times 18000}{360}$$

$$= 900$$

Let's remember!

- Every component of a data is shown by a sector associated with it.
- The measure of the central angle of the sector is in proportion with the number of scores in that component.
- The measure of central angle $(\theta) = \dfrac{\text{Number of scores in component}}{\text{Total number of scores}} \times 360^0$
- A circle of suitable radius should be drawn. Divide the circle in sectors such that the measure of central angle of each sector is proportional to the number of scores in its corresponding component in the data.

Let's learn.

To draw a Pie diagram

We have seen how to read a pie diagram. Now let us learn to draw it.

1. To draw a pie diagram, the whole circle is divided into sectors proportional to the components of the data
2. The measure of central angle of each sector is found by the following formula.

   The measure of central angle of sector θ

$$= \frac{\text{Number of scores in the components}}{\text{Total number of scores}} \times 360$$

A circle of a suitable radius is drawn. Then it is divided into sectors such that, the number of sectors is equal to the number of components in the data.

Let us understand the method through examples.

Solved Examples

Ex. (1) In a bicycle shop, number of bicycles purchased and choice of their colours was as follows. Find the measures of sectors of a circle to show the information by a pie diagram.

Solution : In all 36 bicycles were purchased. Out of them 10 bicycles were white coloured.

$\therefore$ the measure of sector showing white coloured bicycles

$$= \frac{\text{Number of white bicycles}}{\text{Total number of bicycles}} \times 360$$

$$= \frac{10}{36} \times 360 = 100$$

The measures of angles of sector relating to bicycles of other colours can be calculated similarly which are shown in the adjacent table.

Colour	Number of bicycles	Central angle of the sector
White	10	$\frac{10}{36} \times 360^\circ = 100^\circ$
Black	9	$\frac{9}{36} \times 360^\circ = 90^\circ$
Blue	6	60°
Grey	7	70°
Red	4	40°
Total	36	360°

Ex. (2) The following table shows the daily supply of electricity to different places in a town. Show the information by a pie diagram.

Places	Factories	Houses	Roads	Shops	Offices	Others
Supply of electricity (Thousand units)	24	14	7	5	6	4

Solution : The total supply of electricity is 60,000 units. Let us find the measures of central angles and show in the table.

Supply of electricity	Unit	Measure of central angle
Factories	24	$\frac{24}{60} \times 360 = 144^\circ$
Houses	14	$\frac{14}{60} \times 360 = 84^\circ$
Roads	7	$\frac{7}{60} \times 360 = 42^\circ$
Shops	5	$\frac{5}{60} \times 360 = 30^\circ$
Offices	6	$\frac{6}{60} \times 360 = 36^\circ$
Others	4	$\frac{4}{60} \times 360 = 24^\circ$
Total	60	360°

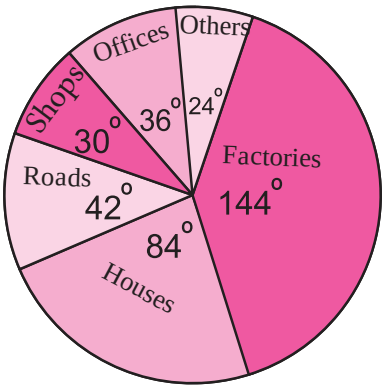

fig 6.12

Steps of drawing pie chart :

(1) As shown in the figure, a circle and a radius is drawn. Then the sectors having measures of angles in the table, (144°,84°, 42°, 30°, 36°, and 24°) were drawn one by one, in the clockwise direction. (While drawing the sectors one by one, we can change their order.)

(2) The components of the data were recorded in the sectors.

Activity :

The monthly expenditure of a family on different items is shown in the following table. Calculate the related central angles and draw a pie chart.

Different items	Percentage of expenditure	Measure of central angle
Food	40	$\frac{40}{100} \times 360 = \square$
Cloting	20	$\square \times \square = \square$
House rent	15	$\square \times \square = \square$
Education	20	$\square \times \square = \square$
Expenditure	05	$\square \times \square = \square$
Total	100	360°

Practice Set 6.6

1. The age group and number of persons, who donated blood in a blood donation camp is given below. Draw a pie diagram from it.

Age group (Yrs)	20-25	25-30	30-35	35-40
No. of persons	80	60	35	25

2. The marks obtained by a student in different subjects are shown. Draw a pie diagram showing the information.

Subject	English	Marathi	Science	Mathematics	Social science	Hindi
Marks	50	70	80	90	60	50

3. In a tree plantation programme, the number of trees planted by students of different classes is given in the following table. Draw a pie diagram showing the information.

Standard	5 th	6 th	7 th	8 th	9 th	10 th
No. of trees	40	50	75	50	70	75

4. The following table shows the percentages of demands for different fruits registered with a fruit vendor. Show the information by a pie diagram.

Fruits	Mango	Sweet lime	Apples	Cheeku	Oranges
Percentages of demand	30	15	25	20	10

5. The pie diagram in figure 6.13 shows the proportions of different workers in a town. Answer the following questions with its help.

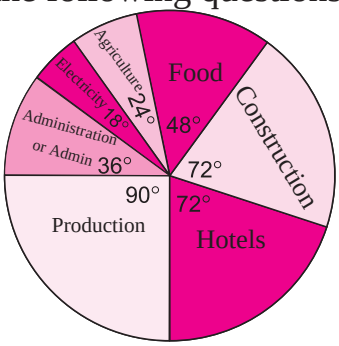

fig 6.13

(1) If the total workers is 10,000; how many of them are in the field of construction?

(2) How many workers are working in the administration?

(3) What is the percentage of workers in production?

6. The annual investments of a family are shown in the adjacent pie diagram. Answer the following questions based on it.

(1) If the investment in shares is ₹ 2000/, find the total investment.

(2) How much amount is deposited in bank?

(3) How much more money is invested in immovable property than in mutual fund?

(4) How much amount is invested in post?

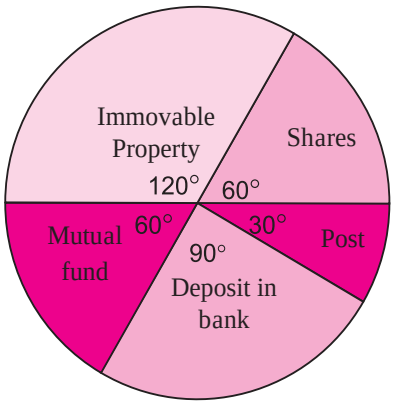

fig 6.14

Miscellaneous Problems – 6

1. Find the correct answer from the alternatives given.

(1) The persons of O- blood group are 40%. The classification of persons based on blood groups is to be shown by a pie diagram. What should be the measures of angle for the persons of O- blood group?

(A) 114° (B) 140° (C) 104° (D) 144°

(2) Different expenditures incurred on the construction of a building were shown by a pie diagram. The expenditure ₹ 45,000 on cement was shown by a sector of central angle of 75°. What was the total expenditure of the construction ?

(A) 2,16,000 (B) 3,60,000 (C) 4,50,000 (D) 7,50,000

(3) Cumulative frequencies in a grouped frequency table are useful to find . . .

(A) Mean (B) Median (C) Mode (D) All of these

(4) The formula to find mean from a grouped frequency table is $\overline{X} = A + \frac{\sum f_i u_i}{\sum f_i} \times h$ In the formula u_i = . . .

(A) $\frac{x_i + A}{h}$ (B) $(x_i - A)$ (C) $\frac{x_i - A}{h}$ (D) $\frac{A - x_i}{h}$

(5)

Distance Covered per litre (km)	12-14	14-16	16-18	18-20
No. of cars	11	12	20	7

The median of the distances covered per litre shown in the above data is in the group . . . . . .

(A) 12-14 (B) 14-16 (C) 16-18 (D) 18-20

(6)

No. of trees planted by each student	1-3	4-6	7-9	10-12
No. of students	7	8	6	4

The above data is to be shown by a frequency polygon. The coordinates of the points to show number of students in the class 4-6 are . . . .

(A) (4, 8) (B) (3, 5) (C) (5, 8) (D) (8, 4)

2. The following table shows the income of farmers in a grape season. Find the mean of their income.

Income (Thousand Rupees)	20-30	30-40	40-50	50-60	60-70	70-80
Farmers	10	11	15	16	18	14

3. The loans sanctioned by a bank for construction of farm ponds are shown in the following table. Find the mean of the loans.

Loan (Thousand rupees)	40-50	50-60	60-70	70-80	80-90
No. of farm ponds	13	20	24	36	7

4. The weekly wages of 120 workers in a factory are shown in the following frequency distribution table. Find the mean of the weekly wages.

Weekly wages (Rupees)	0–2000	2000–4000	4000–6000	6000–8000
No. of workers	15	35	50	20

5. The following frequency distribution table shows the amount of aid given to 50 flood affected families. Find the mean of the amount of aid.

Amount of aid (Thosand rupees)	50–60	60–70	70–80	80–90	90–100
No. of families	7	13	20	6	4

6. The distances covered by 250 public transport buses in a day is shown in the following frequency distribution table. Find the median of the distances.

Distance (km)	200–210	210–220	220–230	230–240	240–250
No. of buses	40	60	80	50	20

7. The prices of different articles and demand for them is shown in the following frequency distribution table. Find the median of the prices.

Price (Rupees)	20 less than	20–40	40–60	60–80	80–100
No. of articles	140	100	80	60	20

8. The following frequency table shows the demand for a sweet and the number of customers. Find the mode of demand of sweet.

Weight of sweet (gram)	0–250	250–500	500–750	750–1000	1000–1250
No. of customers	10	60	25	20	15

9. Draw a histogram for the following frequency distribution.

Use of electricity (Unit)	50–70	70–90	90–110	110–130	130–150	150–170
No. of families	150	400	460	540	600	350

10. In a handloom factory different workers take different periods of time to weave a saree. The number of workers and their required periods are given below. Present the information by a frequency polygon.

No. of days	8–10	10–12	12–14	14–16	16–18	18–20
No. of workers	5	16	30	40	35	14

11. The time required for students to do a science experiment and the number of students is shown in the following grouped frequency distribution table. Show the information by a histogram and also by a frequency polygon.

Time required for experiment (minutes)	20–22	22–24	24–26	26–28	28–30	30–32
No. of students	8	16	22	18	14	12

12. Draw a frequency polygon for the following grouped frequency distribution table.

Age of the donor (Yrs.)	20–24	25–29	30–34	35–39	40–44	45–49
No. of blood doners	38	46	35	24	15	12

13. The following table shows the average rainfall in 150 towns. Show the information by a frequency polygon.

Average rainfall (cm)	0–20	20–40	40–60	60–80	80–100
No. of towns	14	12	36	48	40

14. Observe the adjacent pie diagram. It shows the percentages of number of vehicles passing a signal in a town between 8 am and 10 am

(1) Find the central angle for each type of vehicle.

(2) If the number of two-wheelers is 1200, find the number of all vehicles.

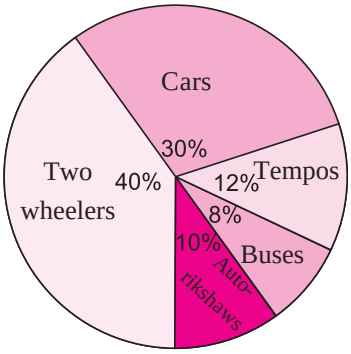

fig 6.15

15. The following table shows causes of noise pollution. Show it by a pie diagram.

Construction	Traffic	Aircraft take offs	Industry	Trains
10%	50%	9%	20%	11%

16. A survey of students was made to know which game they like. The data obtained in the survey is presented in the adjacent pie diagram. If the total number of students are 1000,

(1) How many students like cricket?

(2) How many students like football?

(3) How many students prefer other games?

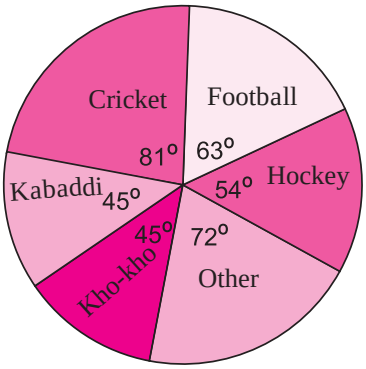

fig 6.16

17. Medical check up of 180 women was conducted in a health centre in a village. 50 of them were short of haemoglobin, 10 suffered from cataract and 25 had respiratory disorders. The remaining women were healthy. Show the information by a pie diagram.

18. On an environment day, students in a school planted 120 trees under plantation project. The information regarding the project is shown in the following table. Show it by a pie diagram.

Tree name	Karanj	Behada	Arjun	Bakul	Kadunimb
No. of trees	20	28	24	22	26

Answers

1. Linear Equations In Two Variables

Practice Set 1.1

2. (1) (2, 4) (2) (3, 1) (3) (6,1) (4) (5, 2)
(5) (-1, 1) (6) (1, 3) (7) (3, 2) (8) (7, 3)

Practice Set 1.2

1. (1)

x	3	**−2**	0
y	**0**	5	3
(x, y)	(3, 0)	**(−2, 5)**	(0, 3)

(2)

x	**4**	−1	0
y	0	**−5**	−4
(x, y)	**(4, 0)**	**(−1,−5)**	(0,−4)

2. (1) (5, 1) (2) (4, 1) (3) (3, -3) (4) (-1, -5) (5) (1, 2.5) (6) (8, 4)

Practice Set 1.3

1. $\begin{vmatrix} 3 & 2 \\ 4 & 5 \end{vmatrix} = 3 \times \boxed{5} - \boxed{2} \times 4 = \boxed{15} - 8 = \boxed{7}$

2. (1) −18 (2) 21 (3) $-\frac{4}{3}$

3. (1) (2, −1) (2) (−2, 4) (3) (3, −2) (4) (2, 6) (5) (6, 5) (6) $(\frac{5}{8}, \frac{1}{4})$

Practice Set 1.4

1. (1) $(\frac{1}{9}, 1)$ (2) (3, 2) (3) $(\frac{5}{2}, -2)$ (4) (1, 1)

Practice Set 1.5

1. The numbers are 5 and 2
Perimeter = 112 unit

2. $x = 12$, $y = 8$, Area = 640 sq. unit,

3. Son's age is 15 years, father's age is 40 years

4. $\frac{7}{18}$

5. A – 30 kg, B – 55 kg

6. 150 km.

Problem Set 1

1. (1) B (2) A (3) D (4) C (5) A

2.

x	−5	$\frac{3}{2}$
y	$-\frac{13}{6}$	0
(x, y)	$(-5, -\frac{13}{6})$	$(\frac{3}{2}, 0)$

3. (1) (3, 2) (2) (-2, -1) (3) (0, 5) (4) (2, 4) (5) (3, 1)

4. (1) 22 (2) −1 (3) 13

5. (1) $(-\frac{2}{3}, 2)$ (2) (1, 4) (3) $(\frac{1}{2}, -\frac{1}{2})$ (4) $(\frac{7}{11}, \frac{116}{33})$ (5) (2, 6)

6. (1) (6, −4) (2) $(-\frac{1}{4}, -1)$ (3) (1, 2) (4) (1, 1) (5) (2, 1)

7. (2) Tea; ₹300 per kg.
sugar; ₹ 40 per kg.
(3) ₹100 notes 20
₹50 notes 10
(4) Manisha's age 23 years
Savita's age 8 years.
(5) Skilled worker's wages ₹ 450.
unskilled worker's wages ₹ 270.
(6) Hamid's speed 50 km/hr.
Joseph's speed 40 km/hr.

2. Quadratic Equations

Practice Set 2.1

1. Any equations of the type $m^2 + 5m + 3 = 0$, $y^2 - 3 = 0$

2. (1), (2), (4), (5) are quadratic equations.

3. (1) $y^2 + 2y - 10 = 0$, $a = 1, b = 2, c = -10$
(2) $x^2 - 4x - 2 = 0$, $a = 1, b = -4, c = -2$
(3) $x^2 + 4x + 3 = 0$, $a = 1, b = 4, c = 3$
(4) $m^2 + 0m + 9 = 0$, $a = 1, b = 0, c = 9$
(5) $6p^2 + 3p + 5 = 0$, $a = 6, b = 3, c = 5$
(6) $x^2 + 0x - 22 = 0$, $a = 1, b = 0, c = -22$

4. (1) 1 is a root, −1 is not. (2) $\frac{5}{2}$ is a root, 2 is not.

5. k = 3 6. k = −7

Practice Set 2.2

1. (1) 9, 6 (2) −5, 4 (3) $-13, -\frac{1}{2}$ (4) $5, -\frac{3}{5}$
(5) $\frac{1}{2}, \frac{1}{2}$ (6) $\frac{2}{3}, -\frac{1}{2}$ (7) $-\frac{5}{\sqrt{2}}, -\sqrt{2}$ (8) $\frac{\sqrt{2}}{\sqrt{3}}, \frac{\sqrt{2}}{\sqrt{3}}$
(9) 25, −1 (10) $-\frac{3}{5}, \frac{3}{5}$ (11) 0, 3 (12) $-\sqrt{11}, \sqrt{11}$

Practice Set 2.3

1. (1) 4,−5 (2) $(\sqrt{6}-1), (-\sqrt{6}-1)$ (3) $\frac{\sqrt{13}+5}{2}, \frac{-\sqrt{13}+5}{2}$

(4) $\frac{\sqrt{2}+2}{3}, \frac{-\sqrt{2}+2}{3}$ (5) $-2, -\frac{5}{2}$ (6) $\frac{2+\sqrt{39}}{5}, \frac{2-\sqrt{39}}{5}$

Practice Set 2.4

1. (1) 1, −7, 5 (2) 2, −5, 5 (3) 1, −7, 0
2. (1) −1, −5 (2) $\frac{3+\sqrt{17}}{2}, \frac{3-\sqrt{17}}{2}$ (3) $\frac{-1+\sqrt{22}}{3}, \frac{-1-\sqrt{22}}{3}$

   (4) $\frac{2+\sqrt{14}}{5}, \frac{2-\sqrt{14}}{5}$ (5) $\frac{-1+\sqrt{73}}{6}, \frac{-1-\sqrt{73}}{6}$ (6) $-1, -\frac{8}{5}$
3. $-\sqrt{3}, -\sqrt{3}$

Practice Set 2.5

1. (1) Roots are distinct and real when $b^2 - 4ac = 5$, not real when $b^2 - 4ac = -5$.

   (2) $x^2 + 7x + 5 = 0$ (3) $\alpha + \beta = 2, \alpha \times \beta = -\frac{3}{2}$
2. (1) 53 (2) −55 (3) 0
3. (1) Real and equal. (2) Real and unequal. (3) Not real.
4. (1) $x^2 - 4x = 0$ (2) $x^2 + 7x - 30 = 0$

   (3) $x^2 - \frac{1}{4} = 0$ (4) $x^2 - 4x - 1 = 0$
5. k = 3 6. (1) 18 (2) 50
7. (1) k = 12 or k = −12 (2) k = 6

Practice Set 2.6

1. 9 years 2. 10 and 12 3. In vertical row 10, in horizontal row 15.
4. Kishor's present age is 10 years and Vivek's present age is 15 years
5. 10 marks 6. No. of pots 6, production cost of each is ₹ 100.
7. 6 km/hr 8. For Nishu 6 days, for Pintu 12 days.
9. Divisor = 9, quotient = 51 10. AB = 7 cm, CD = 15 cm, AD = BC = 5 cm.

Problem Set 2

1. (1) B (2) A (3) C (4) B (5) B (6) D (7) C (8) C
2. (1) and (3) are quadratic equations.

3. (1) −15 (2) 1 (3) 21

4. k = 3 5. (1) $x^2 - 100 = 0$ (2) $x^2 - 2x - 44 = 0$ (3) $x^2 - 7x = 0$

6. (1) Not real. (2) Real and unequal (3) Real and equal

7. (1) $\frac{1+\sqrt{21}}{2}, \frac{1-\sqrt{21}}{2}$ (2) $\frac{1}{2}, -\frac{1}{5}$ (3) 1, −4
(4) $\frac{-5+\sqrt{5}}{2}, \frac{-5-\sqrt{5}}{2}$ (5) Roots are not real. (6) $(2 + \sqrt{7}), (2 - \sqrt{7})$

8. $m = 14$ 9. $x^2 - 5x + 6 = 0$ 10. $x^2 - 4pqx - (p^2 - q^2)^2 = 0$

11. ₹ 100 with Sagar, ₹ 150 with Mukund.

12. 12 and $\sqrt{24}$ or 12 and $-\sqrt{24}$ 13. No. of students 60

14. Breadth 45 m. length 100 m, side of the pond 15 m.

15. For larger tap 3 hours and for smaller tap 6 hours.

3. Arithmetic Progression

Practice Set 3.1

1. (1) Yes, $d = 2$ (2) Yes, $d = \frac{1}{2}$ (3) Yes, $d = 4$ (4) No
(5) Yes, $d = -4$ (6) Yes, $d = 0$ (7) Yes, $d = \sqrt{2}$ (8) Yes, $d = 5$

2. (1) 10, 15, 20, 25, . . . (2) −3, −3, −3, −3, . . . (3) −7, −6.5, −6, −5.5, . .
(4) −1.25, 1.75, 4.75, 7.75, . . .(5) 6, 3, 0, −3 . . . (6) −19, −23, −27, −31

3. (1) $a = 5, d = -4$ (2) $a = 0.6, d = 0.3$ (3) $a = 127, d = 8$ (4) $a = \frac{1}{4}, d = \frac{1}{2}$

Practice Set 3.2

1. (1) $d = 7$ (2) $d = 3$ (3) $a = -3, d = -5$ (4) $a = 70, d = -10$

2. Yes. 121 3. 104 4. 115 5. -121 6. 180

7. 55 8. 55th 9. 60 10. 1

Practice Set 3.3

1. 1215 2. 15252 3. 30450 5. 5040

5. 2380 6. 60 7. 4, 9, 14 or 14, 9, 4 8. −3, 1, 5, 9

Practice Set 3.4

1. ₹ 70455 2. First instalment ₹ 1000, last instalment ₹ 560. 3. ₹ 1,92,000

4. 48, 1242 5. −20°, −25°,−30°,−35°,−40°,−45° 6. 325

Problem Set 3

1. (1) B (2) C (3) B (4) D (5) B (6) C (7) C (8) A (9) A (10) B

2. 40 3. 1, 6, 11, . . . 4. −195 5. 16, -21 6. -1 7. 6, 10

8. 8 9. 67, 69, 71 10. 3, 7, 11, ... , 147 14. ₹ 2000.

4. Financial Planning

Practice Set 4.1

1. CGST 6%, SGST 6% 2. SGST 9%, GST 18%
3. CGST ₹ 784 and SGST ₹ 784
4. The customer gets the belt for ₹ 691.48.
5. Taxable value of toy car is ₹ 1500, CGST ₹ 135, SGST ₹ 135
6. (1) Rate of SGST 14% (2) Rate of GST on AC 28%
   (3) Taxable value of AC ₹ 40,000. (4) Total GST ₹ 11,200.
   (5) CGST ₹ 5600. (6) SGST ₹ 5600.
7. Prasad gets the washing machine for ₹ 48,640 and CGST ₹ 5320, SGST ₹ 5320.

Practice Set 4.2

1. Payable GST ₹ 22,000.
2. Input Tax Credit for Nazama is ₹ 12,500 and her payable GST is ₹ 2250.
3. Ameer Enterprises : Payable GST ₹ 300, payable CGST ₹ 150, payable SGST ₹ 150.
   Akabari Brothers : payable GST ₹ 400, payable CGST ₹ 200, payable SGST ₹ 200.
4. Payable GST ₹ 100 so CGST ₹ 50 and UTGST ₹ 50. 5. CGST = SGST = ₹ 900

Practice Set 4.3

1. (1) MV ₹ 100 (2) FV ₹ 75 (3) At discount of ₹ 5.
2. 25% 3. ₹ 37,040 4. 800 shares
5. Rate of return 5.83% 6. Company A– more profitable.

Practice Set 4.4

1. ₹ 200.60 2. ₹ 999
3.

No. of shares	MV of shares	Total value	Brokerage 0.2%	9% CGST on brokerage	9% SGST on brokerage	Total value of shares
100 B	₹ 45	₹ 4500	₹ 9	₹ 0.81	₹ 0.81	₹ 4510.62
75 S	₹ 200	₹15000	₹ 30	₹ 2.70	₹ 2.70	₹ 14964.60

4. No. of shares sold = 100. 5. Loss of ₹ 8560.

Problem Set 4A

1. (1) C (2) B (3) D (4) B (5) A (6) B
2. Total bill ₹ 28,800 , CGST ₹ 3150, SGST ₹ 3150.

3. ₹ 997.50 4. ₹ 12,500 5. ITC ₹ 4250, payable tax ₹ 250

6. ITC ₹ 1550, payable CGST ₹ 5030, payable SGST ₹ 5030.

7. Taxable value ₹ 75,000, CGST ₹ 4500, SGST ₹ 4500

8. (1) Wholesaler's tax invoice : CGST ₹ 16200; SGST ₹ 16200.
Retailer's tax invoice: CGST ₹ 19,800; SGST ₹ 19,800.
(2) Wholesaler : payable CGST ₹ 2700 and payable SGST ₹ 2700,
Retailer : payable CGST ₹ 3600 and payable SGST ₹ 3600

9. (1) Anna Patil's invoice : CGST ₹ 1960, SGST ₹ 1960
(2) Trader in Vasai : CGST ₹ 2352 and SGST ₹ 2352
(3) Trader in Vasai : payable CGST ₹ 392 and payable SGST ₹ 392

10.

(1)

Person	Payable CGST (₹)	Payable SGST (₹)	Payable GST (₹)
Manufacturer	300	300	600
Distributor	360−300 = 60	60	120
Retailer	390−360 = 30	30	60
Total Tax	390	390	780

(2) Finally, the customer will get the article for ₹ 7280.
(3) Manufacturer to distributor B2B, distributor to retailer B2B,
retailer to customer B2C

Problem Set 4B

1. (1) B (2) B (3) A (4) C (5) A

2. ₹ 130.39 3. 22.2% 4. will get ₹ 21,000 .

5. Will get 500 shares. 6. Profit ₹ 1058.52 7. Company B, as returns are more

8. Will get 1000 shares. 9. ₹ 118.

10. (1) ₹ 1,20,000 (2) ₹ 360 (3) ₹ 64.80 (4) ₹ 120424.80.

11. 1% profit

5. Probability

Practice Set 5.1

1. (1) 8 (2) 7 (3) 52 (4) 11

Practice Set 5.2

1. (1) S = {1H, 1T, 2H, 2T, 3H, 3T, 4H, 4T, 5H, 5T, 6H, 6T} n(S) = 12

(2) S = {23, 25, 32, 35, 52, 53} n(S) = 6

2. S = {Red, Purple, Orange, Yellow, Blue, Green} n(S) = 6

3. S = {Tuesday, Sunday, Friday, Wednesday, Monday, Saturday} n(S) = 6

4. (1) B_1B_2 (2) G_1G_2 (3) B_1G_1 B_2G_1 B_1G_2 B_2G_2

(4) S = $\{B_1B_2, B_1G_1, B_1G_2, B_2G_1, B_2G_2, G_1G_2\}$

Practice Set 5.3

1. (1) S = {1, 2, 3, 4, 5, 6} n(S) = 6

A = {2, 4, 6} n(A) = 3, B = {1, 3, 5} n(B) = 3, C = {2, 3, 5} n(C) = 3

(2) S = {(1,1), . . ., (1, 6), (2,1), . . ., (2, 6), (3, 1), . . .,(3, 6),
(4, 1), . . .,(4,6),(5, 1), . . .,(5, 6),(6, 1), . . .,(6, 6)} n(S) = 36

A = {(1, 5) (2, 4) (3, 3) (4, 2) (5, 1) (6, 6)} n(A) = 6

B = {(4, 6) (5, 5) (5, 6) (6, 4) (6, 5) (6, 6)} n(B) = 6

C = {(1, 1) (2, 2) (3, 3) (4, 4) (5, 5) (6, 6)} n(C) = 6

(3) S = {HHH, HHT, HTT, HTH, THT, TTH, THH, TTT} n(S) = 8

A = {HHH, HHT, HTH, THH} n(A) = 4

B = {TTT} n(B) = 1

C = {HHH, HHT, THH, THT} n(C) = 4

(4) S = {10, 12, 13, 14, 15, 20, 21, 23, 24, 25, 30, 31, 32, 34, 35, 40, 41, 42, 43,
45, 50, 51, 52, 53, 54} n(S) = 25

A = {10, 12, 14, 20, 24, 30, 32, 34, 40, 42, 50, 52, 54} n(A) = 13

B = {12, 15, 21, 24, 30, 42, 45, 51, 54} n(B) = 9

C = {51, 52, 53, 54} n(C) = 4

(5) S = $\{M_1M_2, M_1M_3, M_1F_1, M_1F_2, M_2M_3, M_2F_1, M_2F_2, M_3F_1, M_3F_2, F_1F_2\}$
n(S) = 10

A = $\{M_1F_1, M_1F_2, M_2F_1, M_2F_2, M_3F_1, M_3F_2, F_1F_2\}$ n(A) = 7

B = $\{M_1F_1, M_1F_2, M_2F_1, M_2F_2, M_3F_1, M_3F_2\}$ n(B) = 6

C = $\{M_1M_2, M_1M_3, M_2M_3\}$ n(C) = 3

(6) S = {H1, H2, H3, H4, H5, H6 T1, T2, T3, T4, T5, T6} n(S) = 12

A = {H1, H3, H5} n(A) = 3

B = {H2, H4, H6, T2, T4, T6} n(B) = 6

C = { } n(C) = 0

Practice Set 5.4

1. (1) $\frac{3}{4}$, (2) $\frac{1}{4}$ 2. (1) $\frac{1}{6}$ (2) 0 (3) $\frac{5}{12}$

3. (1) $\frac{7}{15}$ (2) $\frac{1}{5}$ 4. (1) $\frac{4}{5}$ (2) $\frac{1}{5}$ 5. (1) $\frac{1}{13}$ (2) $\frac{1}{4}$

Problem Set – 5

1. (1) B (2) B (3) C (4) A (5) A 2. Vasim's 3. (1) $\frac{1}{11}$ (2) $\frac{6}{11}$
4. $\frac{5}{26}$ 5. (1) $\frac{4}{9}$ (2) $\frac{1}{3}$ (3) $\frac{4}{9}$ 6. $\frac{1}{2}$ 7. (1) $\frac{1}{3}$ (2) $\frac{1}{6}$
8. (1) $\frac{1}{2}$ (2) $\frac{1}{6}$ 9. $\frac{1}{25}$ 10. (1) $\frac{1}{8}$ (2) $\frac{1}{2}$ (3) $\frac{3}{4}$ (4) 1
11. (1) $\frac{5}{6}$ (2) $\frac{1}{6}$ (3) 1 (4) 0 12. (1) $\frac{1}{3}$ (2) $\frac{2}{3}$ (3) $\frac{2}{3}$ 13. $\frac{2}{11}$
14. $\frac{13}{40}$ 15. (1) $\frac{3}{10}$ (2) $\frac{3}{10}$ (3) $\frac{1}{5}$ 16. $\frac{11}{36}$

6. Statistics

Practice Set 6.1

(1) 4.36 hrs (2) ₹ 521.43. (3) 2.82 litre (4) ₹ 35310
(5) ₹ 985 or ₹ 987.5. (6) ₹ 3070 or ₹ 3066.67.

Practice Set 6.2

(1) 11.4 hrs (2) 184.4 means 184 mangoes approximately (3) 74.558 ≈ 75 vehicles
(4) 52750 lamps

Practice Set 6.3

1. 4.33 litre 2. 72 unit 3. 9.94 litre 4. 12.31 years

Practice Set 6.5

1. (1) 60–70 (2) 20–30 and 90–100 (3) 55 (4) 80 and 90 (5) 15

Practice Set 6.6

5. (1) 2000 (2) 1000 (3) 25%
6. (1) ₹ 12000 (2) ₹ 3000 (3) ₹ 2000 (4) ₹ 1000.

Problem Set – 6

1. (1) D (2) A (3) B (4) C (5) C (6) C
2. ₹ 52,500 3. ₹ 65,400 4. ₹ 4250
5. ₹ 72,400 6. 223.13 km. 7. ₹ 32 8. 397.06 gm
14. (1) Cars – 108°, Tempos – 43°, Buses – 29°, Auto-rickshaws – 36°,
Two wheelers – 144°
(2) Total number of vehicles – 3000
16. (1) Cricket – 225, (2) Football – 175 (3) Other games – 200.

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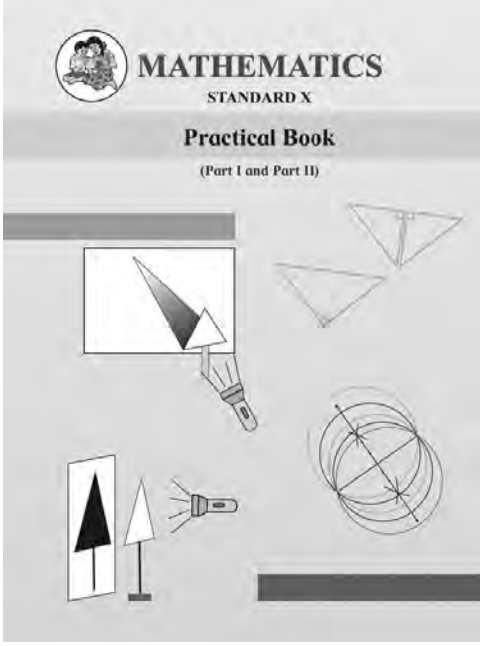

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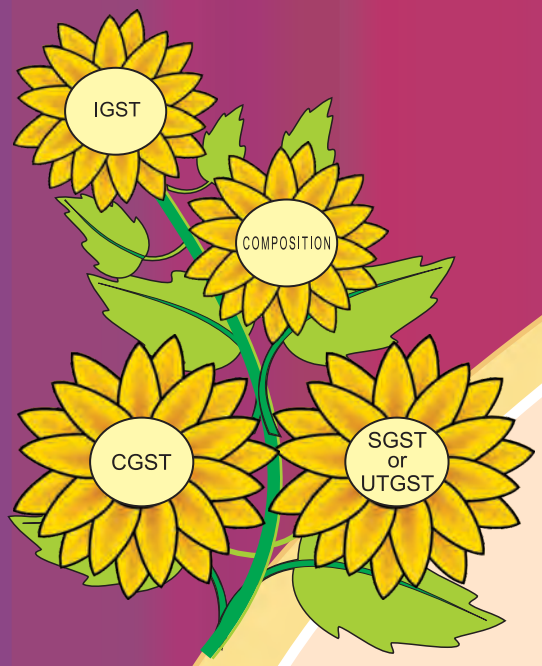

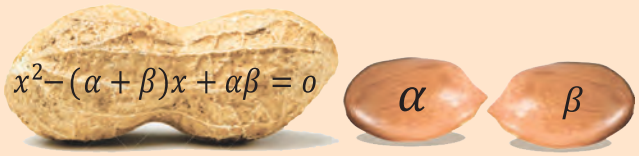

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